fn then_deintegerize_vec

Michael Shoemate (@Shoeboxam)

This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of `then_deintegerize_vec` in `mod.rs` at commit f5bb719 (outdated[1]).

1 Hoare Triple

Precondition

Compiler-verified

- Generic `TO` implements trait `CastInternalRational`

Caller-verified

None

Pseudocode

```
1 def then_deintegerize_vec(k: i32) -> Function[Vec[IBig], Vec[TO]]:
2
3     if k == i32.MIN: #
4         raise ValueError("k must be greater than i32.MIN")
5
6     def element_function(x_i):
7         return TO.from_rational(x_mul_2k(RBig.from_(x_i), k))
8
9     return Function.new(lambda x: [element_function(x_i) for x_i in x])
```

Postcondition

Theorem 1.1. For every setting of the input parameters (`k, TO`) to `then_deintegerize_vec` such that the given preconditions hold, `then_deintegerize_vec` raises an error (at compile time or run time) or returns a valid postprocessor. A valid postprocessor has the following property:

1. (Data-independent errors). For every pair of members x and x' in `input_domain`, `function`(x), `function`(x') either both raise the same error, or neither raise an error.

Proof. By the postcondition of `TO.from_rational`, the outcome of the function is the nearest representable float, and may saturate to positive or negative infinity. The precondition of `x_mul_2k` that k is not `i32.MIN` is satisfied on line 3. Since `TO.from_rational` and `x_mul_2k` are both infallible, the function is infallible, meaning that the function cannot raise data-dependent errors. Therefore the function is a valid postprocessor. □

[1] See new changes with `git diff f5bb719..7b54ec1 rust/src/measurements/noise/nature/float/mod.rs`